Safe Motion Chart - Joint Ranges

Heavy metal 4-DOF arm + gripper | Arduino Uno + PCA9685

File: docs/safe_motion_chart.md

Safe Motion Chart · Calibrated Values (2026-10-23)

Arm: heavy metal 4-DOF + gripper (MG995/MG996R joints)

Controller: Arduino Uno + PCA9685 (I2C, 50 Hz)

Purpose: keep the arm stable, avoid servo overload, and complete a pick-and-place cycle without tipping.

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1. Channel map and calibrated limits

Joint	PCA9685 ch	Full calibrated range	Operational clamp used in code	Notes
Base / waist	**0**	200·500	**335·415**	center 375; rotate only while arm is upright
Shoulder	**1**	200·340	**250·275**	center **260 = vertical**; stays vertical during reach
Elbow	**2**	200·420	**205·310**	center **300 = vertical**; does the reach down
Wrist pitch	**6**	300·450	**370·380**	fixed at **375** during motion
Wrist rotation	**4**	·	·	**SERVO DEAD · not used**
Gripper	**7**	120·200	**120 open / 200 closed**	do not command outside this travel
PCA9685 ch3	3	·	·	**FAULTY OUTPUT · never use**

> The full calibrated range is wider than what the assembled heavy arm can safely use. The code uses the narrower operational clamp above.

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2. Why the arm was collapsing

The shoulder was doing the deep bend. That hangs the entire arm's weight in front of the base. The shoulder servo could not hold that torque, the arm fell, and lifting it back from the extended position was impossible.

Fix: the shoulder stays essentially vertical (258). The **elbow** does the reach down (300 · 278 · 248 · 218), so only the light forearm moves. Lifting reverses the sequence in small steps (218 · 240 · 262 · 284 · 300). The base rotates only after the arm is folded upright.

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3. Pick-and-place sequence (implemented)

Step	Base	Shoulder	Elbow	Gripper	Purpose
Recover / home	375	260	300	open 120	compact, balanced
1. Reach down	375	258	300 · 278 · 248 · **218**	open 120	elbow-only reach
2. Grip	375	258	218	**closed 200**	grasp object
3. Lift	375	260	218 · 240 · 262 · 284 · **300**	closed	light forearm lifts first
4. Rotate	375 · bin	260	300	closed	base moves only while upright
5. Lower	bin	258	300 · 278 · 248 · **218**	closed	lower over bin
6. Release	bin	258	218	**open 120**	drop object
7. Lift	bin	260	218 · · · **300**	open	return upright
8. Home	375	260	300	open	ready for next cycle

Bin angles: RED = **335**, GREEN = **375**, BLUE = **415** (all inside the 335·415 base clamp).

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4. Safety rules

1. **Clamp the base to the table** before running.
2. **Fold the arm upright before every base rotation.**
3. **Move one joint at a time** with delays between steps.
4. **Every pulse is clamped** in firmware; never command outside the limits above.
5. **Support the arm during the first run** and during the 6-second startup recovery.
6. **Power:** XL4016 5·6 V, ·6 A peak, common ground, 2200 µF capacitor across PCA9685 V+/GND.

7. **Payload:** start with ·100 g; maximum intended ·180 g.
8. **No full-range sweeps** on the assembled arm.

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5. Known hardware faults

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| Fault | Symptom | Action |
|---|---|---|
| PCA9685 channel 3 | no output on any servo | avoid channel 3 |
| Wrist-rotation servo (ch4) | holds but never rotates, even with bare shaft | replace the servo |
| Gripper travel | only 120·200 ticks move | use open 120 / closed 200 |
| Shoulder deep bend | arm falls and cannot lift | use elbow-reach mode above |
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6. Related files

- Firmware: `code/pick_and_place/pick_and_place.ino`
- Integrated sensors: `code/integrated_sorting/integrated_sorting.ino`
- Wiring: `hardware/pin_connections.md`
- Sensor guide: `docs/sensor_integration.md`